Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name:** Binary Minds

b. **Team leader name:** Mohammed Iliyas, Rinkle Agrawal

c. **Problem Statement:** Every release, product managers have to make sure nothing breaks CCPA rules, but nobody can check every guideline by hand. So a dark pattern slips into the live site, and the company ends up with a fine and a bad name.

Brief about the idea

! The problem

- India's CCPA banned 13 dark patterns in 2023, including drip pricing, basket sneaking and false urgency.
- Enforcement is live: Zepto fined ₹7 lakh, McAfee ₹1 lakh; in June 2025 all e-commerce platforms were told to self-audit.
- Audits are periodic, manual, time consuming while checkout and pricing flows release continuously.
- A flow can drift out of compliance with the next release, unnoticed until someone else spots it.

✓ The solution

- Tests your live store like a real customer would, before every single release.
- Flags each dark pattern with a screenshot and the exact CCPA rule it breaks.
- Suggests a fix, re-tests the store, and shows the new compliance score.
- Saves every run, so you always have proof of compliance ready to show.

5
AI agents, one pipeline

<=50 Lacs
Savings from penalty for violation

5 of 13
CCPA dark patterns tested currently

1
Single click pipeline run to test compliance

Opportunities

Why now: CCPA banned 13 dark patterns (2023) and enforcement is live; in June 2025 it directed all e-commerce platforms to self-audit. Penalties go up to ₹10 lakh, and ₹50 lakh for a repeat contravention.

1 How different is it from existing ideas?

- Consultants and page scanners audit periodically, one page or screenshot at a time. Dark patterns hide between pages.
- Verdicts rest on the official CCPA text and live data (“Only 3 left!” vs 178 in stock).
- A plain AI chat prompt can give one opinion with no systematic output. TrustLens runs the same journey every release, so results can be compared between versions and audit.

2 How will it solve the problem?

- TrustLens plays the full customer journey in a real browser and compares checkpoints, identifying dark patterns.
- Proves each finding with screenshots and a cited CCPA rule.
- Generate a full complaince report covering every pipeline stage
- Stores each run as a trail compliance and product teams can audit.

3 USP of the proposed solution

- Prove, cite, and suggests fix in one pipeline, not five separate tools. .
- Grounded verdicts: official CCPA text retrieved at run time, not left to model memory.
- Two fixes per finding: a statutory change with code fix, and a conversion-friendly alternative.
- Repeatable by design: the same scripted journey every run, so release-to-release scores are fair and consistent.

List of features offered by the solution

1

Agent Dashboard

Pick the version, start a run and watch the five stages finish live. Select respective agent for screenshots, findings and reasoning.

2

Customer journey simulation

An agent shops your live store like a real customer, from catalogue to cart to checkout to cancellation, capturing a screenshot at every checkpoint.

3

Specified Clauses

Every ruling quotes the CCPA 2023 definition and legal basis, with a confidence level for human review.

4

Dual fixes

A compliance change with code fix, paired with an ethical, conversion-friendly alternative.

5

Run comparison

Two runs side by side post fix implementation with before and after comparison and the score change.

6

Compliance trail

Score history across releases, and a downloadable PDF report.

7

Jira Ticket hand-off

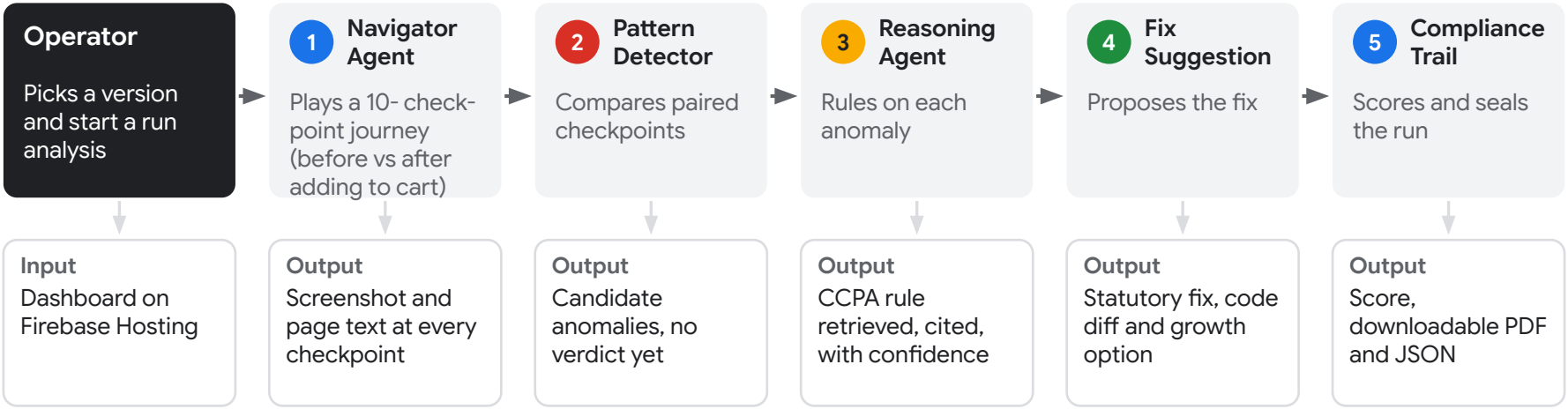
One click drafts a Jira issue pre-filled with the clause, evidence and fix.

8

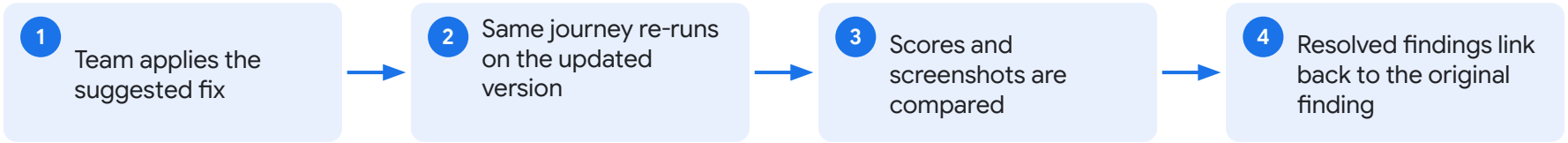
Dark Pattern Guide

All 13 CCPA patterns with excerpts and enforcement precedents; the 5 tested today are marked.

Process flow diagram or Use-case diagram

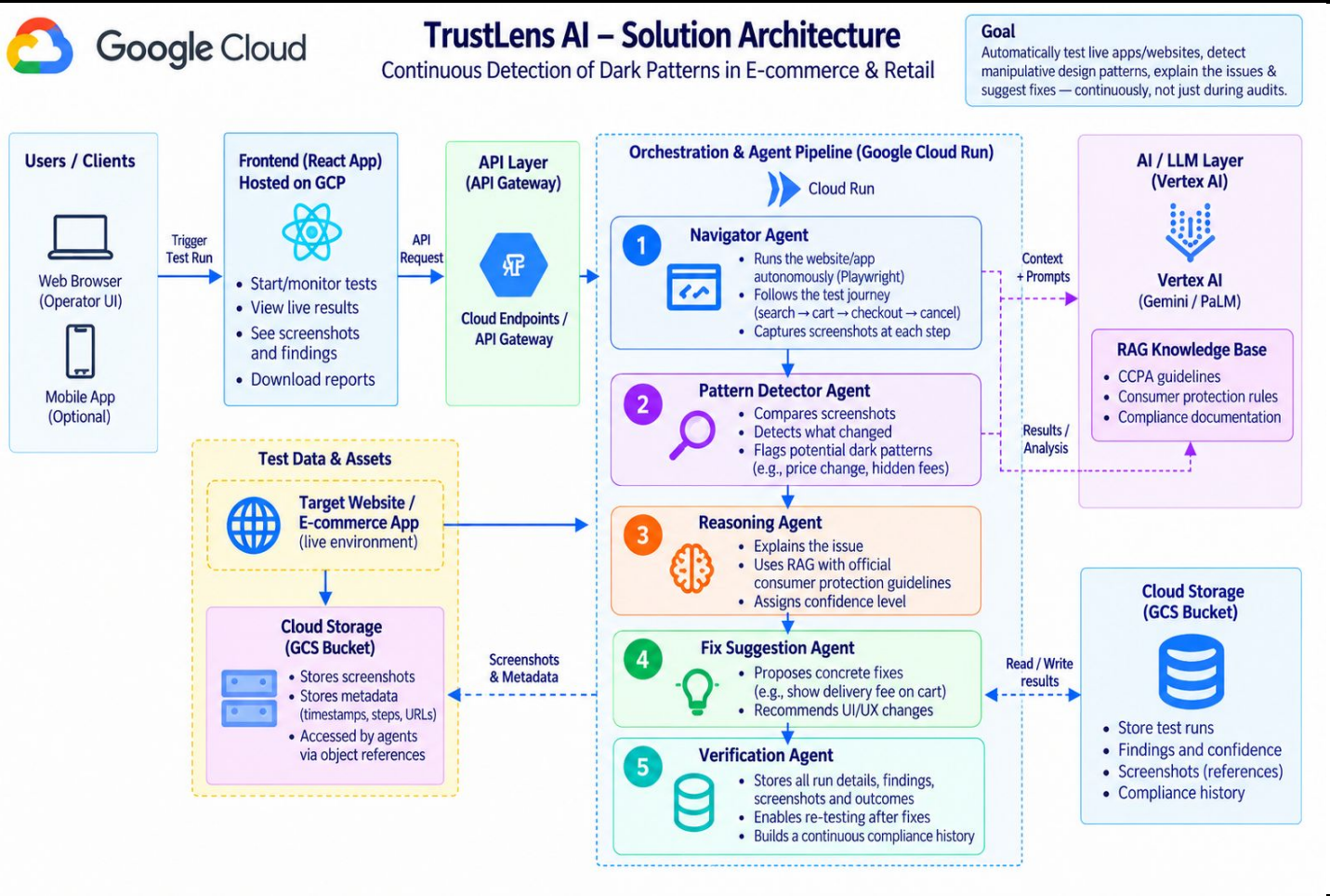


After every release: the re-test loop



Users: compliance and product teams, as a pre-release check on dark patters.

Solution Architecture



Technologies used in the solution

Google Cloud

Cloud Run
FastAPI backend and Navigator job

Gemini on Agent Platform (Vertex AI)
Vision, reasoning, suggestion fixes

Cloud Storage
Evidence and CCPA corpus, runs, findings, trail, inventory data

Artifact Registry
Storing the docker image

IAM + Secret Manager
Service Account, access and secrets

Application stack

React
Operator dashboard

FastAPI (Python)
Backend and orchestration

Playwright + Chromium
Scripted customer journey

PDF + JSON
Report and certificate

On the roadmap

Cloud Scheduler
Release-pipeline re-tests

BigQuery
Analytics across platforms

Jira, ServiceNow
Live ticket sync

Gemini exploration
Navigation without a script

Theme: Retail & Commerce, intelligent customer and business experiences, built on Google Cloud.

Estimated implementation cost

Service	Cost driver	Expected at prototype scale	Per run cost	100 runs per month
Cloud Run	API requests and one Navigator job per run	Low: short jobs, scales to zero	under $0.01, mostly inside free tier	negligible
Gemini on Vertex AI	~10 screenshots plus text per run; reasoning and fix calls	Main variable cost; grows with runs	about $0.05 (about ₹4 to 5)	about $5
Cloud Storage	Screenshots, PDF reports, CCPA corpus	Low: small files per run	negligible	negligible
IAM + Secret Manager	Service accounts and secrets	Negligible	negligible	negligible

No fixed cost

No licences or servers to run: every service above is serverless and pay-per-use.

Indicative only

Validate with the Google Cloud pricing calculator once run volume and storefront count are fixed.

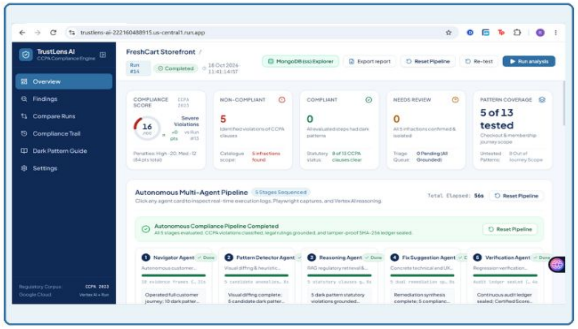
Snapshots of the prototype

TrustLens AI: product screens

Run > Detect > Cite > Fix > Re-test > Report

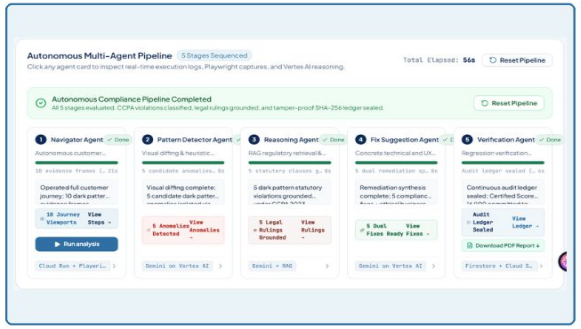
1 Run console

Score, violation counts and 5-agent pipeline in one view



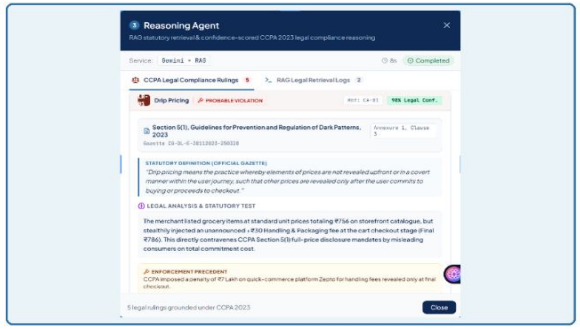
2 Agent pipeline

Five agents, each with its output, timing and Google Cloud service



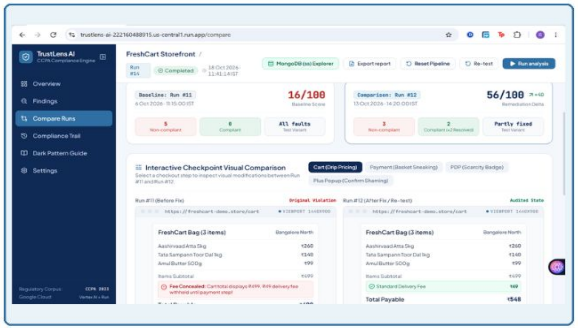
3 Cited CCPA ruling

Official clause, legal analysis, precedent and confidence



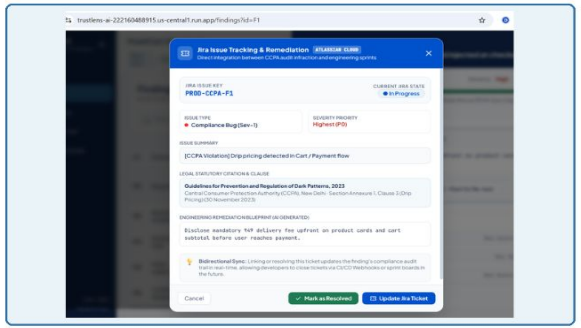
4 Run comparison

Before vs after fix, side by side, with score change



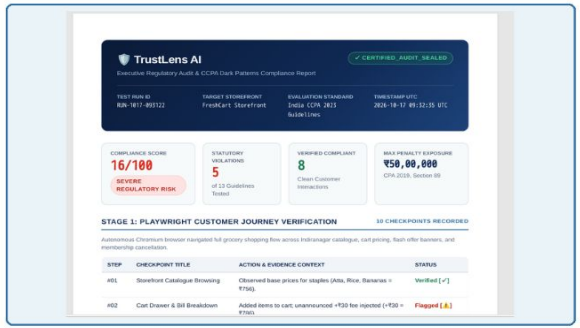
5 Jira hand-off

One-click ticket with clause, evidence and engineering fix



6 Compliance report

Sealed audit report covering every pipeline stage



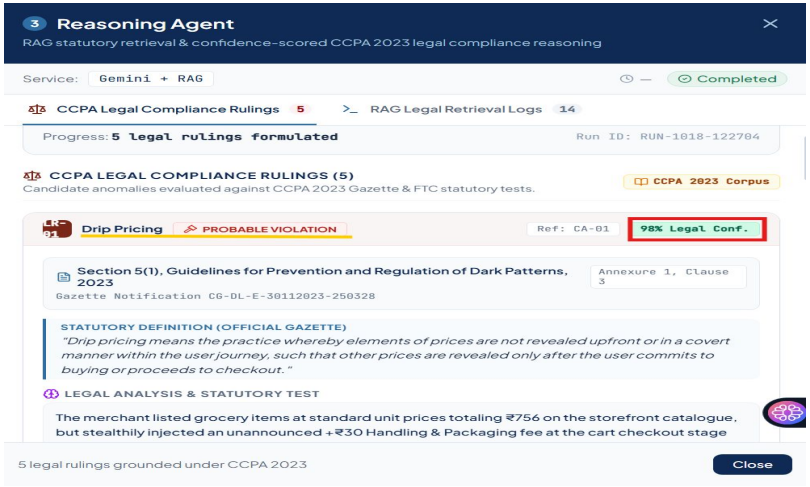
Prototype Performance report/Benchmarking

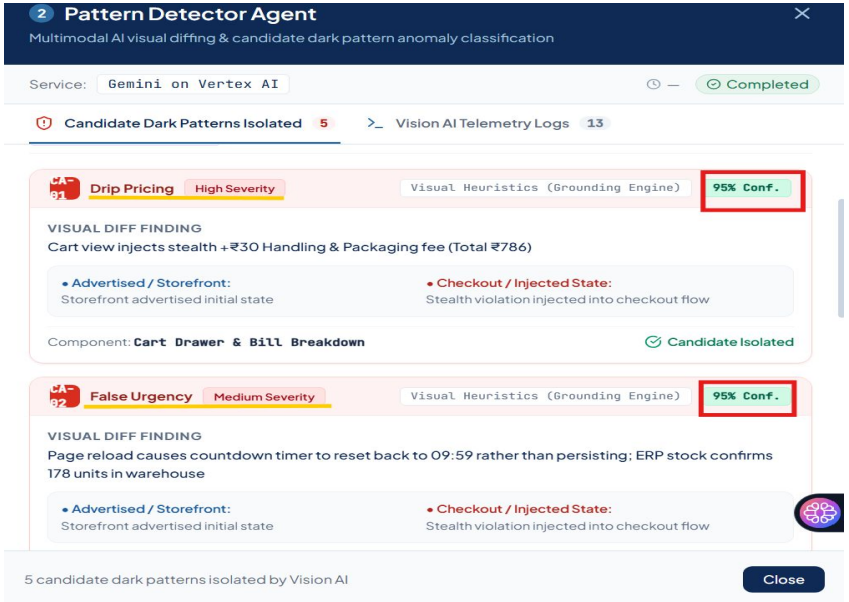
40s – 2 min

- End-to-end run time: about 40 s on our demo, up to about 2 min for larger sites

92–95%

- Pattern Detector confidence when flagging candidate dark patterns





96–98%

- Reasoning Agent legal confidence on each CCPA ruling

5 of 13

- CCPA dark patterns tested across a 10-checkpoint customer journey

Future Development Scope

Additional details: scope and limits

- **Tested scope only.** 5 of 13 patterns on one scripted journey. A 100/100 score means no violations found in that scope.
- **Decision support.** Verdicts carry a confidence level for human review. Penalty figures are statutory ceilings (up to ₹10 lakh; ₹50 lakh for repeat), not predicted fines.
- **Demo integrations.** The Jira hand-off and the inventory feed use mock data, not live Jira, ServiceNow or warehouse systems.

Future development

- **More patterns.** Interface interference, nagging, trick questions and the rest of the 13 CCPA patterns.
- **Continuous runs.** Cloud Scheduler plus a release-pipeline hook, so every deploy is re-tested.
- **Live integrations.** Jira and ServiceNow ticket sync, and real inventory feeds.
- **Exploration mode.** Gemini-driven navigation for sites without a scripted journey.
- **Scale and reuse.** BigQuery analytics across platforms; other rulebooks (EU Digital Services Act, US click-to-cancel) and sectors such as travel, fintech and subscriptions.

Provide links to your:

- GitHub Public Repository: https://github.com/rinkle-agarwal/AI-Builder-Hackathon
- Demo Video Link (3 Minutes 26 seconds): https://drive.google.com/file/d/1I8Gp-BKTOZwSVJREOPX0J63peJjQ8ac8/view?usp=sharing
- Final Product Link: https://trustlens-ai-222160488915.us-central1.run.app/

How judges can test the demo (no login needed)

1 Open the product link

The demo has no authentication or login: open the link and the dashboard loads.

2 Choose a storefront

Run the test storefront with dark patterns, then the fixed version. The demo html stores can be selected after the run analysis button. Both test HTML files are in the repo as well.

3 Start a run

Watch the five stages, open the findings, compare the two runs and download the output report.

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Thank you

